# Equivalent Analog Filtering

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# Key Concepts

1) One way to filter a continuous-time or analog signal is to sample the signal, apply a discrete-time system to the samples, and reconstruct a continuous-time signal.

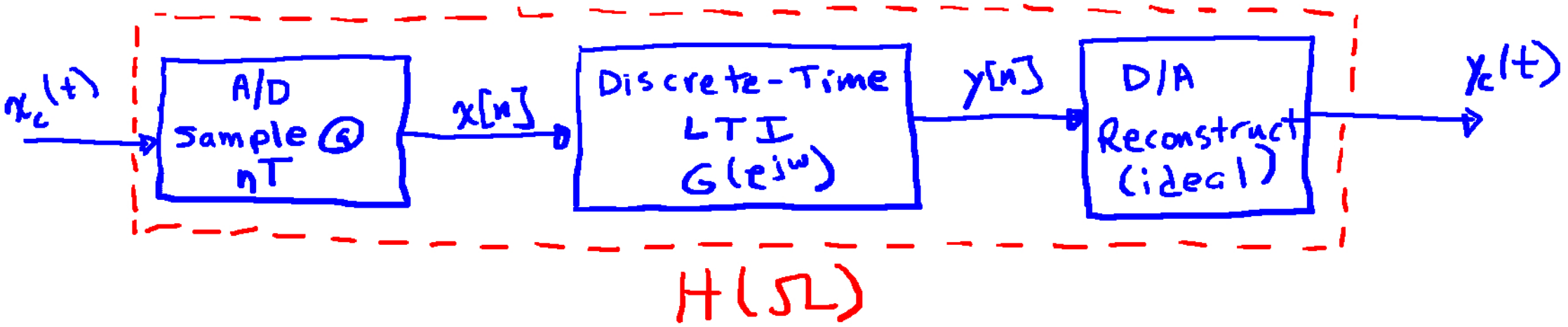
2) A practical question is how to choose the discrete-time system frequency response $G(e^{j\omega})$ so that the equivalent analog filter $H(\Omega)$ represented by sampling, discrete-time filtering, and reconstruction implements a desired characteristic.

3) If the conditions of the sampling theorem are satisfied and ideal low pass filtering is used for reconstruction, then $H(\Omega) = G(e^{j\Omega T})$.

4) Critical continuous-time frequencies $\Omega_i$ such as edges of filter pass bands in $H(\Omega)$ map directly to critical discrete-time frequencies $\omega_i = \Omega_i T$ in $G(e^{j\omega})$.

# Video Screenshots

# Equivalent Analog Filtering




Desired filter $H(j\Omega) \Rightarrow G(e^{j\omega})$ ?

Example: Speech - remove low-frequency noise

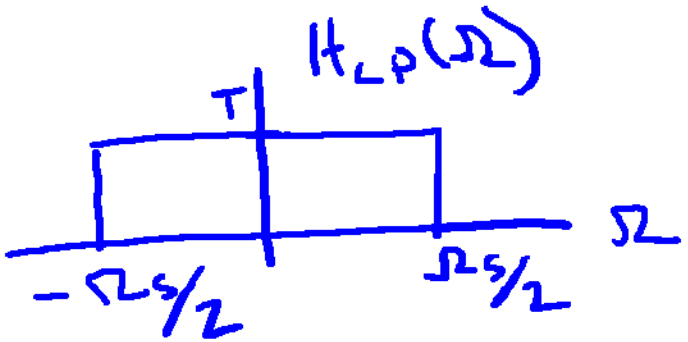
$|\Omega| < 100\text{Hz}$ Assume $X_c(j\Omega) = 0$ $|\Omega| > 8000\pi$ $\frac{\text{rads}}{\text{sec}}$

$\Omega_s = 20000\pi$ rads/sec

Analysis: $x_c(t) \xleftrightarrow{FT} X_c(\Omega)$ ; $x[n] \xleftrightarrow{DTFT} X(e^{j\omega})$

$y_c(t) \xleftrightarrow{FT} Y_c(\Omega)$ ; $y[n] \xleftrightarrow{DTFT} Y(e^{j\omega})$

Recall - $\omega = \Omega T$ ; ideal reconst.




$$Y_c(\Omega) = H_{LP}(\Omega)\, Y(e^{j\omega})\Big|_{\omega = \Omega T}$$

$$= H_{LP}(\Omega)\, Y(e^{j\Omega T})$$

$$Y(e^{j\omega}) = G(e^{j\omega})\, X(e^{j\omega})$$

$$Y_c(\Omega) = H_{LP}(\Omega)\, G(e^{j\Omega T})\, X(e^{j\Omega T})$$

Since $X_{cs}(\Omega) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c(\Omega - k\Omega_s)$

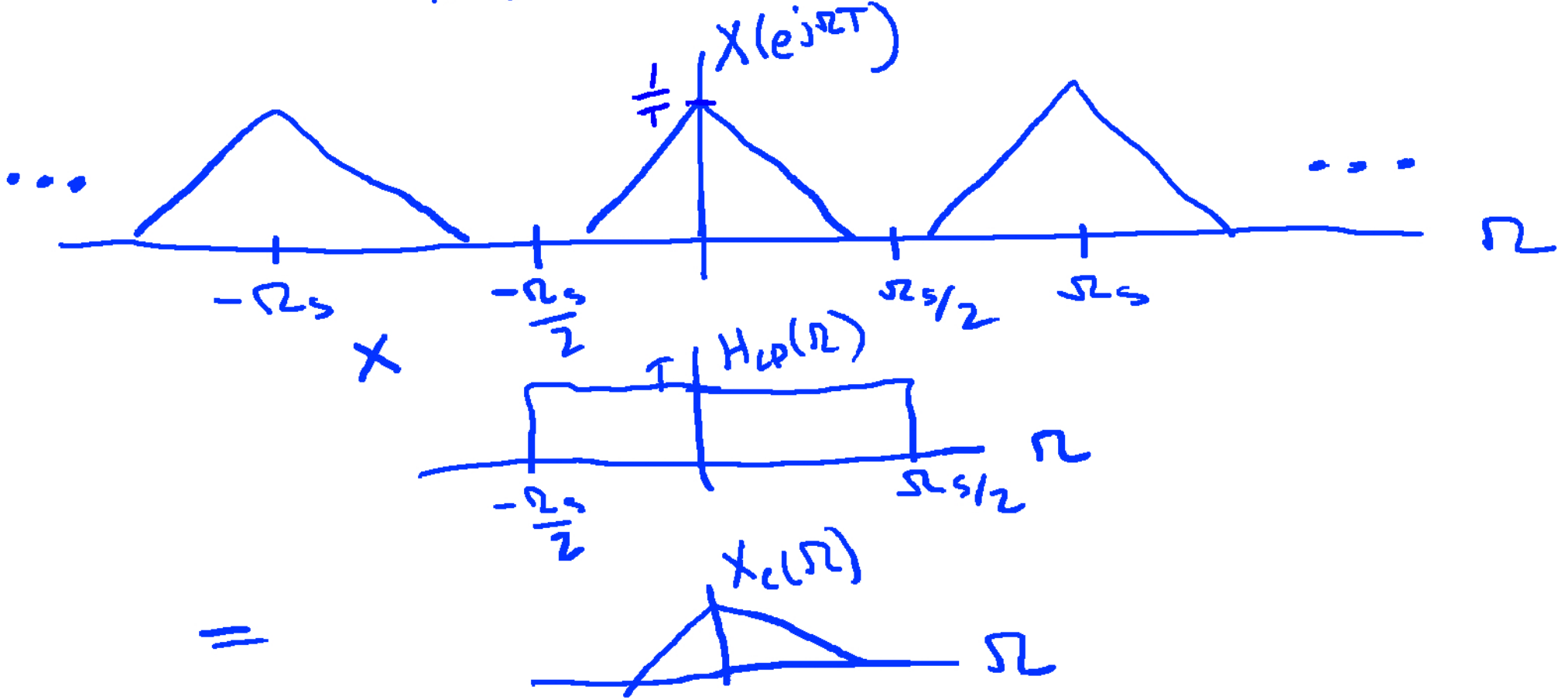
$$X(e^{j\omega}) = X_{cs}(\Omega)\Big|_{\Omega=\frac{\omega}{T}} = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c\left(\frac{\omega}{T} - k\Omega_s\right)$$

$$X(e^{j\Omega T}) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c(\Omega - k\Omega_s)$$

$$\underbrace{Y_c(\Omega)}_{\text{output}} = H_{LP}(\Omega)\, G(e^{j\Omega T})\, \frac{1}{T}\sum_{k=-\infty}^{\infty} \underbrace{X_c(\Omega - k\Omega_s)}_{\text{input}}$$

Assuming $|X_c(\Omega)| = 0, |\Omega| > \frac{\Omega_s}{2}$ (Sampling Thm)

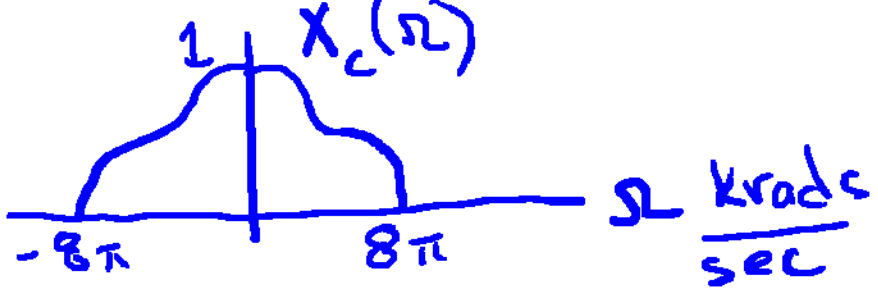
$$H_{LP}(\Omega) \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(\Omega - k\Omega_s) = X_c(\Omega)$$



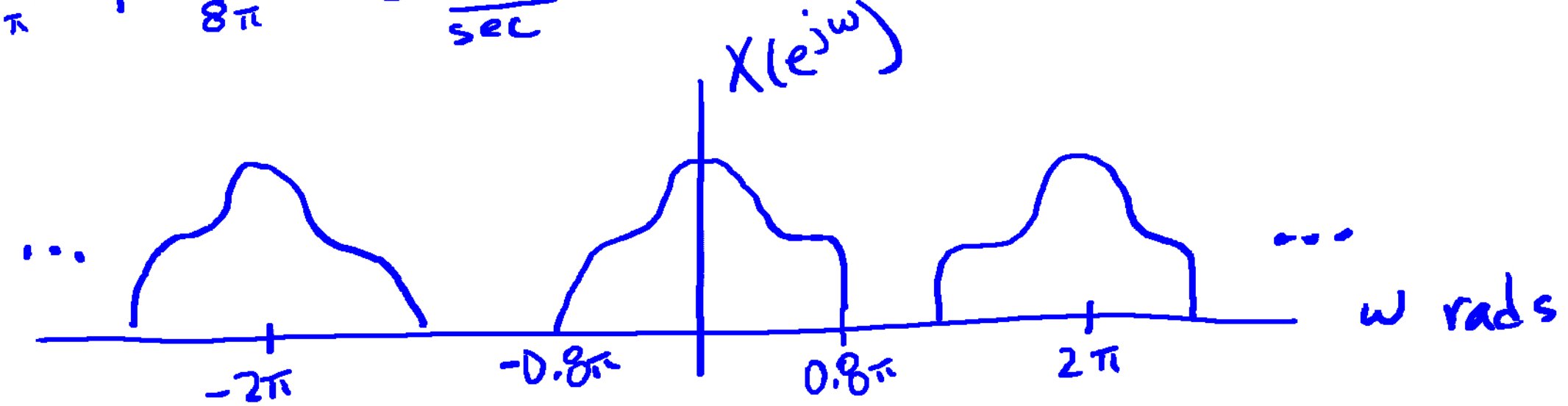
Thus $Y_c(\Omega) = G(e^{j\Omega T})\, X_c(\Omega)$ $\quad |\Omega| < \Omega_s/2$

Specs on $H(\Omega)$ transfer to $G(e^{j\Omega T})$ for $|\Omega| < \Omega_s/2$

Speech Example: $f_s = 10^4$ Hz $\Rightarrow T = 10^{-4}$ sec, $\Omega_s = 20\pi \frac{\text{Krads}}{\text{sec}}$

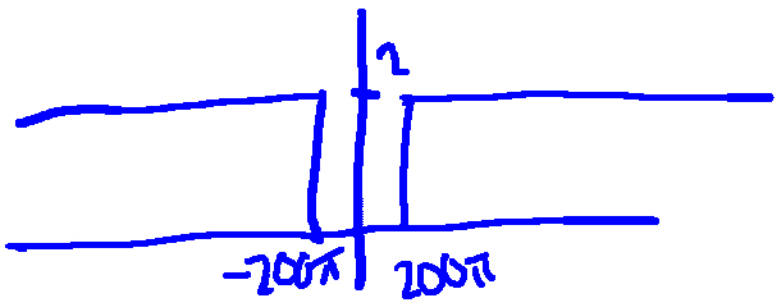



$\omega = \Omega\,(10^{-4})$

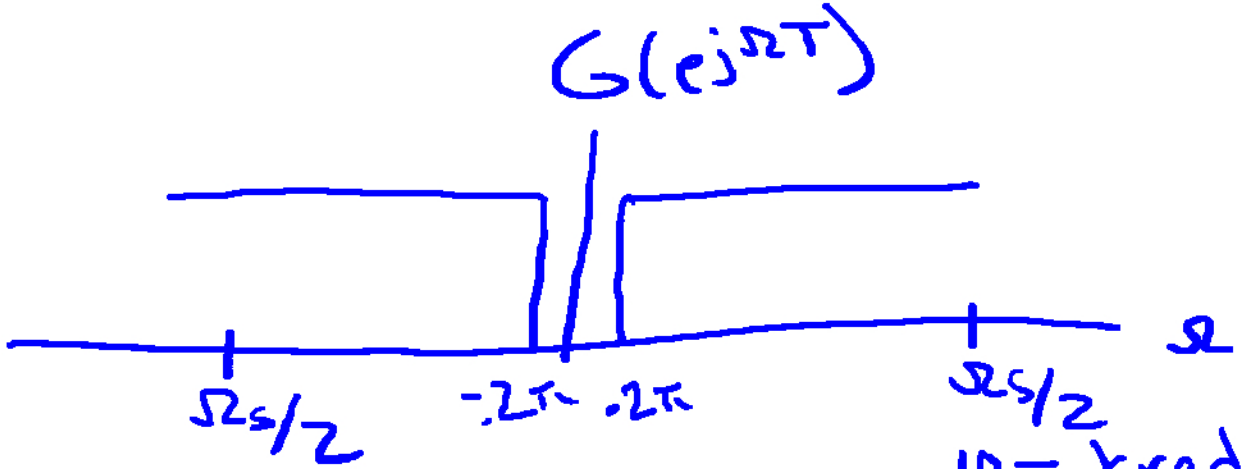


Highpass filter w/ cutoff 100 Hz ⇒ $200\pi \frac{\text{rads}}{\text{sec}}$

$H(\Omega)$



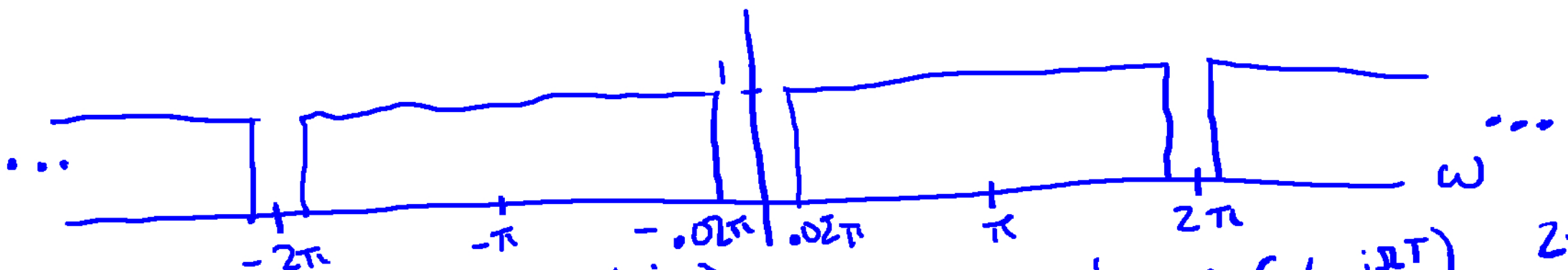

$G(e^{j\Omega T})$




$10\pi$ krads/sec

$\omega = \Omega T$

$G(e^{j\omega})$




$G(e^{j\omega})$ is $2\pi$ periodic ⇒ $G(e^{j\Omega T})$ $\frac{2\pi}{T} = \Omega_s$

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